Proposition 13: no linear bound for the spectral distance

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport* (file `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository). Statement, equation, page and line numbers refer to this PDF; `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

The folder `GsimOT/Proposition14_SpectralLowerBound` formalises Proposition 13 with the construction of App. F.1, Lemma 49, Theorem 3 (4) and the remark after Proposition 13.

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Main statement in Lean. `GsimOT.Prop14.proposition13`: for every $r \geq 1$, with $n = |V(G_r)| = 4r^2 + 3r + 1$,

$$d_2(G_r, H_r) = \sqrt{r} \qquad \text{and} \qquad \tfrac{1}{2\sqrt{2}} \leq d_2^{\mathrm{OT}}(G_r, H_r) \leq 1;$$

`GsimOT.Prop14.proposition14` adds $\frac{1}{2\sqrt{2}} \leq \frac{r}{\sqrt{n}} \leq d_2^{\mathrm{OT}}(G_r, H_r)$ (Lemma 49). `GsimOT.Prop14.no_power_bound`: for every $\alpha > \frac{3}{4}$ and C, some G_r, H_r violate $\delta_2 \leq C(\delta_2^{\mathrm{OT}})^{\alpha}$.

Correspondence

Paper	Page	Statement	Lean declaration(s) and location
Proposition 13	11	$d_2(G_r, H_r) = \sqrt{r}$ and $\frac{1}{2\sqrt{2}} \leq d_2^{\mathrm{OT}}(G_r, H_r) \leq 1$, $n = 4r^2 + 3r + 1 = \Theta(r^2)$	`GsimOT.Prop14.proposition13` `Proposition14_SpectralLowerBound/Proposition14.lean:46`
Prop. 13 + Lemma 49	11, 45	the same with $\frac{1}{2\sqrt{2}} \leq \frac{r}{\sqrt{n}} \leq d_2^{\mathrm{OT}}(G_r, H_r)$, the bound $\frac{r}{\sqrt{n}} \leq d_2^{\mathrm{OT}}(G_r, H_r)$ being Lemma 49	`GsimOT.Prop14.proposition14` `Proposition14_SpectralLowerBound/Proposition14.lean:25`
Theorem 3 (4)	4	$\delta_2(G_r, H_r) \geq \sqrt{r} \cdot \delta_2^{\mathrm{OT}}(G_r, H_r)$	`GsimOT.Prop14.theorem4_item4` `Proposition14_SpectralLowerBound/Proposition14.lean:81`

after Prop. 13	11	No bound $\delta_2 \leq C_\alpha(\delta_2^{\mathrm{OT}})^{\alpha}$ for $\alpha > \frac{3}{4}$ (l. 391–393)	`GsimOT.Prop14.no_power_bound`
			`Proposition14_SpectralLowerBound/Proposition14.lean:91`
			`GsimOT.Prop14.deltaPQ_eq`
			`Proposition14_SpectralLowerBound/Proposition14.lean:60`
			`GsimOT.Prop14.deltaOTPQ_bounds`
			`Proposition14_SpectralLowerBound/Proposition14.lean:67`

Paper	Page	Statement	Lean declaration(s) and location

App. F.1, Fig. 7	45	Core $X = \{u\} \cup \{v_{i,k}\}$, clocks C, tags T_i; $N(v_{i,k}) = T_i \cup \{c_1, \ldots, c_k\}$; u isolated in G_r, adjacent to C in H_r	`GsimOT.Prop14.Core`
			`Proposition14_SpectralLowerBound/Construction.lean:27`
			`GsimOT.Prop14.Marker`
			`Proposition14_SpectralLowerBound/Construction.lean:30`
			`GsimOT.Prop14.Vtx`
			`Proposition14_SpectralLowerBound/Construction.lean:33`
			`GsimOT.Prop14.graph`
			`Proposition14_SpectralLowerBound/Construction.lean:61`
			`GsimOT.Prop14.Gr`
			`Proposition14_SpectralLowerBound/Construction.lean:67`
			`GsimOT.Prop14.Hr`
			`Proposition14_SpectralLowerBound/Construction.lean:70`

App. F.1	45	$\lvert V\rvert = n = 4r^2 + 3r + 1$; H_r has minimum degree $\geq r$	`GsimOT.Prop14.card_Vtx`
			`Proposition14_SpectralLowerBound/Construction.lean:35`
			`GsimOT.Prop14.le_deg_Hr`
			`Proposition14_SpectralLowerBound/Construction.lean:138`

App. F.1	45	$d_2(G_r, H_r) = \sqrt{r}$: row of $\pi(u)$ (spectral norm $\geq$ row norm) and the star for id (l. 1412–1416)	`GsimOT.Prop14.distPQ_eq`
			`Proposition14_SpectralLowerBound/SpectralDistance.lean:120`
			`GsimOT.Prop14.sqrt_row_le_opNorm22`
			`Proposition14_SpectralLowerBound/SpectralDistance.lean:47`
			`GsimOT.Prop14.sqrt_le_opNorm_alignedDiff`
			`Proposition14_SpectralLowerBound/SpectralDistance.lean:73`
			`GsimOT.Prop14.opNorm_diff_refl_le`
			`Proposition14_SpectralLowerBound/SpectralDistance.lean:101`

Lemma 49	45	$\frac{r}{\sqrt{n}} \leq d_2^{\mathrm{OT}}(G_r, H_r) \leq 1$	`GsimOT.Prop14.lemma53` `Proposition14_SpectralLowerBound/Lemma53.lean:420`

Proof of Lemma 49	45–46	Shifts σ_i and $Q = \frac{1}{2r} \sum_i P_{\sigma_i}$	`GsimOT.Prop14.shift`
			`Proposition14_SpectralLowerBound/Lemma53.lean:113`
			`GsimOT.Prop14.tau`
			`Proposition14_SpectralLowerBound/Lemma53.lean:129`
			`GsimOT.Prop14.Qs`
			`Proposition14_SpectralLowerBound/Lemma53.lean:137`
			`GsimOT.Prop14.Qs_isDS`
			`Proposition14_SpectralLowerBound/Lemma53.lean:170`

Paper	Page	Statement	Lean declaration(s) and location
Proof of Lemma 49	46	Vectors d_{xy}; $C_2(Q) \le 1$ (tags: Cauchy–Schwarz, clocks orthogonal; l. 1424–1436)	GsimOT.Prop14.dv Proposition14_SpectralLowerBound/Lemma53.lean:184 GsimOT.Prop14.sum_qc_dvDot_le Proposition14_SpectralLowerBound/Lemma53.lean:239 GsimOT.Prop14.opNorm_Bq_le Proposition14_SpectralLowerBound/Lemma53.lean:386 GsimOT.Prop14.dOT_le_one Proposition14_SpectralLowerBound/Lemma53.lean:400
Proof of Lemma 49	46	Test vector $\mathbf{1}_R$: $(B\mathbf{1}_R)(x,y) = \mathbf{1}_R(x,y)(\deg_G x - \deg_H y)$, hence $C_2(R) \ge r/\sqrt{n}$ (l. 1437–1446)	GsimOT.Prop14.le_opNorm_Bmat Proposition14_SpectralLowerBound/Lemma53.lean:50 GsimOT.Prop14.le_dOT_of_deg Proposition14_SpectralLowerBound/Lemma53.lean:101 GsimOT.Prop14.le_dOT Proposition14_SpectralLowerBound/Lemma53.lean:407
Proof of Prop. 13	46	$\frac{r}{\sqrt{n}} \ge \frac{1}{2\sqrt{2}}$ since $8r^2 \ge n$ (l. 1447–1448)	GsimOT.Prop14.proposition14 Proposition14_SpectralLowerBound/Proposition14.lean:25

Shared notions used (from GsimOT/Common)

Paper	Page	Statement	Lean declaration(s) and location
Sec. 1, App. A	2–3, 14	Couplings $\mathrm{DS}^*(V,W)$; x^a with $0^a := 0$ (convention $0^0 = 0$)	GsimOT.IsCoupling Common/OTDistances.lean:36 GsimOT.powZ Common/OTDistances.lean:42
Sec. 1, Eqs. (1), (2)	2–3	$B(G,H,Q)$ and $B_{p\to q}(G,H,Q)$	GsimOT.B Common/OTDistances.lean:58 GsimOT.Bmat Common/OTDistances.lean:70

Paper	Page	Statement	Lean declaration(s) and location

Eq. (2), Sec. 2	3, 5	$\delta^{\mathrm{OT}}_{p\to q}$ (couplings) and $d^{\mathrm{OT}}_{p\to q}$ ($\mathrm{DS}(V,W)$), both infima	`GsimOT.deltaOTPQ`
			`Common/OTDistances.lean:74`
			`GsimOT.dOT`
			`Common/OTDistances.lean:79`

Secs. 1–2, Apps. A–B	3, 5, 14–15	$\delta^{\mathrm{OT}}_{p\to q} = n^{-\kappa_{p\to q}} d^{\mathrm{OT}}_{p\to q}$ for graphs of the same order n ($d^{\mathrm{OT}}_{p\to q}$ is the unnormalised version, $s(n) = n^{\kappa_{p\to q}}$), via $\mathrm{DS}(V,W) = n\,\mathrm{DS}^*(V,W)$ and the homogeneity of $B_{p\to q}$ and of $\|\cdot\|_{p\to q}$	`GsimOT.deltaOTPQ_eq_dOT`
			`Common/OTDistances.lean:131`
			`GsimOT.Bmat_smul`
			`Common/OTDistances.lean:117`
			`GsimOT.opNorm_smul_of_nonneg`
			`Common/OTDistances.lean:94`

App. A	14–15	"All minima … are attained" for $\delta^{\mathrm{OT}}_{p\to q}$ and $d^{\mathrm{OT}}_{p\to q}$: lower semicontinuity of $Q \mapsto \|B_{p\to q}(G,H,Q)\|$, compactness	`GsimOT.lowerSemicontinuous_opNorm_Bmat`
			`Common/Attainment.lean:136`
			`GsimOT.isCompact_setOf_isCoupling`
			`Common/Attainment.lean:168`
			`GsimOT.exists_deltaOTPQ_eq`
			`Common/Attainment.lean:199`
			`GsimOT.exists_dOT_eq`
			`Common/Attainment.lean:223`

Notes on the formalisation

1. **Upper bound of Lemma 49.** The paper computes $C_2(Q)^2 = \|K^\top K\|_2 = 1$ for the block form $\begin{pmatrix} 0 & K \\ K^\top & 0 \end{pmatrix}$. The formalisation instead shows $\sum_{(x,y)} Q(x,y)(d_{xy}\cdot\zeta)^2 \le \|\zeta\|^2$. It uses Cauchy–Schwarz on each tag block and the orthogonality of the clock markers, and concludes $|y^\top B x| \le 1$ through Lemma 19. This is the same estimate, organised as a bilinear bound.

2. The lower bound of Lemma 49 is proved for arbitrary graphs (`le_dOT_of_deg`: a vertex whose degree differs from every degree in H by $\ge c$ forces $d_2^{\mathrm{OT}} \ge c/\sqrt{n}$) and then specialised to $c = r$.

3. The remark after Proposition 13 is formalised as `no_power_bound`. It uses $\delta_2(G_r,H_r) = \sqrt{r}/n$, $\delta_2^{\mathrm{OT}}(G_r,H_r) \le 1/n$ and $n \le 8r^2$; the lower estimate $\delta_2^{\mathrm{OT}} = \Omega(n^{-1})$ is not needed for this conclusion.

4. Indices: $v_{i,k}$ is `some (i, k)` and u is `none` in `Core`; clocks and tags are the two summands of `Marker`. G_r and H_r are one construction `graph` with a Boolean flag for the edges at u.

5. `proposition13` is Proposition 13. `proposition14` states Proposition 13 together with the bound $\frac{r}{\sqrt{n}} \leq d_2^{\mathrm{OT}}(G_r, H_r)$ of Lemma 49, which the proof of Proposition 13 combines with $\frac{r}{\sqrt{n}} \geq \frac{1}{2\sqrt{2}}$; `proposition13` is derived from it.